COMPARATIVE TANGKHUL: LEXICAL AND PHONOLOGICAL RECONSTRUCTIO

Authors: David R. Mortensen, Jennifer Keogh

Journal/Venue: UC Berkeley Sino-Tibetan Linguistic Monograph Series | Year: 2003
Subject: Comparative Tibeto-Burman Linguistics & Dialectology | Language: English & Tangkhul

ABSTRACT:

This paper presents a preliminary lexical and phonological reconstruction of Proto-Tangkhul (PTk), the ancestral mother language of the Tangkhulic branch within the Tibeto-Burman family. Based on extensive field data collected across Ukhrul (Standard Tangkhul), Huishu, Kachai, East Tusom, and Phadang, we establish regular sound correspondences between onset stops, liquid clusters (*pl-, *kl-, *pr-, *kr-), and vowel developments. We demonstrate that Standard Tangkhul underwent lenition of syllable codas while peripheral varieties like Huishu and Tusom preserved archaic conservative distinctions.

Keywords: Proto-Tangkhul, Phonology, Dialectology, Kachai, Huishu, Ukhrul, Sound Shifts

1. INTRODUCTION & GENETIC SUBGROUPING

The Tangkhulic languages, spoken primarily in Ukhrul and Kamjong districts of Manipur and adjoining regions of Myanmar, constitute an independent, deeply conservative branch of Tibeto-Burman. Grierson (1903) categorized them under the generic rubric 'Kuki-Chin-Naga', but subsequent historical work (Matisoff 1972, Burling 2003) confirmed their discrete genetic status. Standard Tangkhul (Ukhrul) functions as the literary standard and regional lingua franca, yet individual villages preserve distinct daughter languages with varying degrees of mutual intelligibility.

Tangkhul Linguistic Data & Analysis:

* **kapha (Ukhrul) / ge-pha (Kachai) : good, beautiful**
  Analysis: Voicing contrast in prefixes across dialects.
* **ngalanao (Ukhrul) / nga-la-na (Huishu) : maiden, daughter**
  Analysis: Preservation of proto-stem *nga-la.
* **mayarnao (Ukhrul) / ma-ja-na (Tusom) : young man, son**
  Analysis: Palatal glide development from proto-rhotic *r.

2. PHONOLOGICAL CORRESPONDENCES & SYLLABLE STRUCTURE

Proto-Tangkhul possessed a sesquisyllabic or disyllabic structure *Ce-CVC. Modern Standard Tangkhul preserves the unstressed nominalizing/adjectival prefix *ke- (written ka-) and negative *me- (written ma-). In Kachai, these prefixes undergo secondary voicing assimilation (ge- / be-). In coda position, Proto-Tangkhul *-(k, t, p) are realized as unreleased glottal stops or weak stops in Ukhrul, whereas final *-l is lost in Ukhrul but preserved in Huishu.

Tangkhul Linguistic Data & Analysis:

*** ngashan (Ukhrul) / ga-shan (Phadang) : clan, lineage, custom**

Analysis: Retroflex sibilant development in western dialects.

*** tui (Ukhrul) / tol (Huishu) : word, speech**

Analysis: Loss of final liquid *-l in standard Ukhrul.

*** meikazat (Ukhrul) / mi-za (Kachai) : fire flame**

Analysis: Sesquisyllabic prefix reduction.

3. TONAL CATEGORIES & PITCH CONTOURS

Proto-Tangkhul tones correlate systematically with syllable coda sonorancy and proto-voicing. Standard Ukhrul distinguishes three level tones: High (55), Mid (33), and Low (22). Stems ending in original voiceless stops condition High abrupt pitch, whereas sonorant codas allow tonal register contrasts that interact with morphological suffixation.

Tangkhul Linguistic Data & Analysis:

* **lui (High tone) : paddy field, terraced plot**
  Analysis: High level register.
* **lui (Mid tone) : to take back, fetch**
  Analysis: Mid level lexical tone.
* **lui (Low tone) : song, chant melody**
  Analysis: Low register tonal contour.

REFERENCES & BIBLIOGRAPHY

* Burling, Robbins. 2003. The Tibeto-Burman languages of Northeastern India. The Sino-Tibetan Languages, 169-191.

* Grierson, George A. 1903. Linguistic Survey of India, Vol. III, Part II: Bodo, Naga and Kachin Groups. Calcutta.

* Matisoff, James A. 1972. Tangkhul Naga and comparative Tibeto-Burman. Tonogenesis in Southeast Asia, 271-283.

* Mortensen, David R. 2012. Database of Tangkhulic Languages. UC Berkeley Phonology Lab.

Digital Archiving Notice:
Digitized and integrated into the Tangkhul Language Library Open Research Corpus.

A RECONSTRUCTION OF PROTO-TANGKHULIC RHYMES

Authors: David R. Mortensen, James A. Miller

Journal/Venue: Linguistics of the Tibeto-Burman Area (LTBA) | Year: 2013
Subject: Historical Phonology & Sino-Tibetan Rhymes | Language: English & Tangkhul

ABSTRACT:

We reconstruct the rhyme system of Proto-Tangkhulic (PTk) based on a comparative corpus of over 800 cognate sets spanning Standard Tangkhul (Ukhrul), Huishu, Kachai, Champhung, and Tusom. We demonstrate that Proto-Tangkhulic distinguished phonemic vowel length before dental and velar codas (*-a:t vs *-at; *-a:k vs *-ak), and retained the Proto-Tibeto-Burman lateral coda *-l, which merged into front glides or open vowels in Standard Tangkhul. This reconstruction resolves longstanding anomalies in the external comparison of Tangkhul with Jingpho and Lolo-Burmese.

Keywords: Rhymes, Proto-Tangkhulic, Tibeto-Burman, Vowel Length, Liquid Codas, LTBA

1. THE PTK RHYME INVENTORY

The reconstructed rhyme inventory of Proto-Tangkhulic includes 24 distinct rhymes with stop and nasal codas: *-am, *-an, *-ang, *-ap, *-at, *-ak, *-a:t, *-a:k, *-im, *-in, *-ing, *-ip, *-it, *-ik, *-um, *-un, *-ung, *-up, *-ut, *-uk, and lateral rhymes *-al, *-il, *-ul. While Standard Tangkhul regularly neutralizes vowel length before codas, the conservative Huishu dialect preserves qualitative differences that verify the length distinction.

Tangkhul Linguistic Data & Analysis:

*** kapik (Ukhrul) / pi:k (Huishu) : to sleep**

Analysis: Reflects PTk *Ce-pi:k (long vowel before velar stop).

*** kashak (Ukhrul) / sa' (Kachai) : to eat**

Analysis: PTk *Ce-syak (short rhyme with glottal realization).

*** tui (Ukhrul) / tul (Huishu) : speech, language**

Analysis: PTk *to:l (preservation of final lateral in Huishu).

2. EVOLUTION OF LATERAL CODAS (*-L)

One of the major breakthroughs of Proto-Tangkhulic historical phonology is recognizing the regular fate of PTB *-l. In Standard Ukhrul, PTB *-l vocalized to a high front offglide [-i] after back vowels (e.g. *tul > tui 'speech'; *ngul > ngui 'silver/money'). In Huishu, *-l is preserved intact as an alveolar lateral continuant [-l], proving that lateral loss in Ukhrul was a secondary innovation of the central dialect group.

Tangkhul Linguistic Data & Analysis:

*** ngui (Ukhrul) / ngul (Huishu) : silver, coin, money**

Analysis: Direct cognate with Written Tibetan dngul.

*** kahui (Ukhrul) / hul (Huishu) : to bark (of dog)**

Analysis: Reflects PTB *hwel.

*** malui (Ukhrul) / melol (Huishu) : to forget**

Analysis: Preservation of liquid coda in archaic peripheral varieties.

3. IMPLICATIONS FOR SINO-TIBETAN SUBGROUPING

Recognizing the conservative nature of Proto-Tangkhulic rhymes provides a crucial benchmark for the genetic classification of the Indo-Burma borderlands. Tangkhulic shares specific rhyme mergers with Maring and Kuki-Chin, yet preserves distinct prefixal morphology that establishes it as a primary coordinate branch of Tibeto-Burman.

Tangkhul Linguistic Data & Analysis:

*** ngalei (Ukhrul) : earth, ground**

Analysis: PTk *C-lay cognate with Proto-Kuki-Chin *lay.

*** shing (Ukhrul) : wood, tree**

Analysis: PTk *sing cognate with Written Burmese sac and Tibetan shing.

REFERENCES & BIBLIOGRAPHY

* Benedict, Paul K. 1972. Sino-Tibetan: A Consortium. Cambridge University Press.

* Matisoff, James A. 2003. Handbook of Proto-Tibeto-Burman: System and Philosophy of Sino-Tibetan Reconstruction. UC Press.

* Mortensen, David R. & James A. Miller. 2013. A Reconstruction of Proto-Tangkhulic Rhymes. LTBA 36(1): 1-28.

Digital Archiving Notice:

Digitized and integrated into the Tangkhul Language Library Open Research Corpus.

TANGKHUL NAGA AND COMPARATIVE TIBETO-BURMAN

Authors: Prof. James A. Matisoff

Journal/Venue: Sino-Tibetan Etymological Dictionary and Thesaurus (STEDT) Monograph | Year: 1972
Subject: Comparative Sino-Tibetan Linguistics | Language: English & Tangkhul

ABSTRACT:

This foundational paper analyzes the historical phonology and lexical morphology of Tangkhul Naga in the context of Proto-Tibeto-Burman (PTB) reconstruction. We examine over 120 basic vocabulary items from Pettigrew (1918) and Bhat (1969), identifying regular sound laws linking Tangkhul velar prefixes (ke-), nasal prefixes (me-), and sibilant causatives (se-) to pan-Sino-Tibetan morphosyntactic patterns. Tangkhul provides indispensable evidence for the antiquity of Tibeto-Burman sesquisyllabicity and nominalizing prefixation.

Keywords: STEDT, Matisoff, Proto-Tibeto-Burman, Cognates, Prefixation, Etymology

1. THE CRUCIAL ROLE OF TANGKHUL IN PTB RECONSTRUCTION

Tangkhul Naga occupies a uniquely illuminating position in Tibeto-Burman comparative linguistics. While spoken in relative geographical proximity to the Meithei (Manipuri) kingdom and the Kuki-Chin hills, its grammatical structure preserves ancient prefixal paradigms that have eroded in more isolating languages of Southeast Asia. Rev. William Pettigrew's 1918 grammar and dictionary provided early linguists with their first systematic view of this archaic system.

Tangkhul Linguistic Data & Analysis:

*** kathei : to see / perception**

Analysis: PTB *m-ti-k; prefix ka- preserves nominalized state.

*** kashak : to eat / food**

Analysis: PTB *dzya; regular correspondence with Written Tibetan za-ba.

*** kathi : to die / death**

Analysis: PTB *s-tey; pan-Tibeto-Burman root found from Burma to Himalayas.

2. PREFIXAL MORPHOLOGY AND DERIVATIONAL FUNCTIONS

In Tangkhul, verbal roots rarely occur bare in citation form. Instead, they carry the ubiquitous prefix ka-, which functions simultaneously as an infinitive marker, action nominalizer, and adjectivalizer (e.g., pha 'be good' -> kapha 'good'). Negative clauses replace this with the discontinuous morpheme ma-...-mana. Causatives are derived through suffixal -sak or archaic stem-initial aspiration, directly parallel to the Tibetan s- causative prefix.

Tangkhul Linguistic Data & Analysis:

*** kapha (good) / makapha-mana (not good) : Affirmative vs Negative**

Analysis: Discontinuous negation frame.

*** sasak (cause to do) eina kasa (to do) : Causative derivation**

Analysis: Causative auxiliary affixation.

3. TANGKHUL ETYMOLOGICAL COGNATES IN STEDT

In the Sino-Tibetan Etymological Dictionary and Thesaurus (STEDT), Tangkhul cognates verify roots for anatomical terms, celestial phenomena, and agricultural implements. The preservation of distinction between initial velars and labiovelars corroborates reconstructions in Jingpho, Garo, and Old Chinese.

Tangkhul Linguistic Data & Analysis:

*** mik : eye**

Analysis: PTB *s-myak; identical coda stop with Burmese myak.

*** kakhut : foot, leg**

Analysis: PTB *r-kay; velar stop preservation.

*** kho / khong : stream, ditch**

Analysis: PTB *k-l(w)ang.

REFERENCES & BIBLIOGRAPHY

* Bhat, D. N. S. 1969. Tankhur Naga Vocabulary. Deccan College, Poona.

* Matisoff, James A. 1972. Tangkhul Naga and comparative Tibeto-Burman.
STEDT Paper 1, UC Berkeley.

* Pettigrew, William. 1918. Tangkhul Naga Grammar and Dictionary.
Shillong: Government Press.

Digital Archiving Notice:

Digitized and integrated into the Tangkhul Language Library Open Research Corpus.

TANGKHUL-NAGA GRAMMAR: A STUDY OF WORD FORMATION AND MORPH

Authors: Dr. Victor Ahum

Journal/Venue: Manipur University Academic Monograph Series | Year: 1997
Subject: Descriptive Linguistics & Morphology | Language: English & Tangkhul

ABSTRACT:

This study provides the first comprehensive modern morphological analysis of Standard Tangkhul Naga by a native linguist. Focusing on word-formation strategies, the dissertation examines the typology of roots, derivational prefixes, inflections, and compound formations. We investigate nominal compounding (endocentric and exocentric), reduplication, verbal valency alterations, and postpositional case affixation, demonstrating how Tangkhul balances agglutinating morphology with analytic syntax.

Keywords: Word Formation, Morphology, Affixation, Compounding, Valence, Ahum, Ukhrul

1. MORPHOLOGICAL TYPOLOGY & ROOT CLASSES

Tangkhul word formation operates primarily through agglutination with strict suffixal dominance in nominal inflection and prefixal dominance in verbal derivation. Monosyllabic lexical roots fall into three major syntactic categories: Noun roots (shim 'house', ram 'country'), Verb roots (sa 'do', thei 'see', zat 'go'), and Deictic/Particle bases (hi 'this', chi 'that'). Nominal roots can occur bare as free morphemes, whereas verbal roots generally require affixes or compounding.

Tangkhul Linguistic Data & Analysis:

*** shim (house) -> shimkhar (village community) -> shimsak (construction) : Nominal root extension**

Analysis: Root compounding.

*** tui (word) -> tuitam (lesson) -> tuingashik (covenant) : Discourse compounding**

Analysis: Semantic broadening.

2. NOMINAL COMPOUNDS & GENDER MARKERS

Compounding is the most productive lexical expansion mechanism in Tangkhul. Compounds exhibit head-initial and head-final constructions. Natural gender in animate nouns is marked not by grammatical inflection but by sex-specifying enclitics: -va for feminine (e.g. siva 'bitch', harva 'hen') and -mayar for masculine (simayar 'male dog', harmayar 'rooster').

Tangkhul Linguistic Data & Analysis:

*** mayarnao (man/son) / ngalanao (woman/daughter) : Human gender terms**

Analysis: Root + gender enclitic + diminutive nao.

*** haiku (cooking vessel) -> meiku (fire basin) : Utensil compounding**

Analysis: Endocentric nominal compound.

3. VALENCE MODIFICATION & ASPECTUAL INFLECTION

Tangkhul verbal roots undergo valence modification through prefixation and auxiliary compounding. Reflexives take prefix nge- (ngathang 'wash oneself'), reciprocals take nge- with root reduplication (ngazek-ngazek 'converse mutually'), and causatives take suffix -sak. Completive aspect is marked by -ser, durative by -kapam, and future intentional by -ra.

Tangkhul Linguistic Data & Analysis:

*** ana i-li theisaku : He caused me to see (showed me)**

Analysis: Causative suffix -sak.

*** athum ngayeiserie : They all love one another completely**

Analysis: Completive aspect -ser.

REFERENCES & BIBLIOGRAPHY

* Ahum, Victor. 1997. Tangkhul-Naga Grammar: A Study of Word Formation. Ph.D. dissertation, Manipur University.

* Comrie, Bernard. 1989. Language Universals and Linguistic Typology. University of Chicago Press.

* Pettigrew, William. 1918. Tangkhul Naga Grammar and Dictionary. Shillong: Assam Secretariat Press.

Digital Archiving Notice:
Digitized and integrated into the Tangkhul Language Library Open Research Corpus.

ASPIRATION DISSIMILATION AND PHONOLOGICAL RULES IN TANGKHUL NA

Authors: Deepthi Gopal

Journal/Venue: Carnegie Mellon University Working Papers in Linguistics | Year: 2016
Subject: Phonetics & Laboratory Phonology | Language: English & Tangkhul

ABSTRACT:

In Tangkhul Naga, prefixal stop consonants exhibit systematic aspiration dissimilation reminiscent of Grassmann's Law in Indo-European. When a verbal root begins with an aspirated stop (ph-, th-, kh-), the preceding nominalizing/infinitive prefix surfaces as an unaspirated stop [ke-]. Conversely, when the verbal root begins with an unaspirated stop or sonorant, the prefix is realized with robust aspiration [khe-] in conservative varieties. Using acoustic measurements of Voice Onset Time (VOT) and burst spectra from 8 native speakers, this paper confirms that aspiration is a syntagmatically governed feature bounded by the phonological word.

Keywords: Aspiration, Dissimilation, Grassmann's Law, Phonology, VOT, Acoustics

1. THE PHENOMENON OF ASPIRATION DISSIMILATION

Phonologists have long studied long-distance consonant interactions such as consonant harmony and dissimilation. In Tangkhul, the interaction between the prefix consonant and the root-initial consonant exhibits a classic Obligatory Contour Principle (OCP) effect for the feature [spread glottis]. Two adjacent syllables within the same phonological stem cannot both surface with contrastive aspiration.

Tangkhul Linguistic Data & Analysis:

*** ke-pha [kapha] : good, beautiful**

Analysis: Root initial aspirated [ph-]; prefix is unaspirated [ke-].

*** khe-pa [khapa] : to read / recite**

Analysis: Root initial unaspirated [p-]; prefix surfaces with aspiration [khe-].

*** ke-thi [kathi] : to die**

Analysis: Root [th-]; prefix dissimilates to unaspirated stop.

2. ACOUSTIC METHODOLOGY & VOICE ONSET TIME (VOT)

Acoustic recordings of 120 prefix-root words were analyzed in Praat. Burst duration, Voice Onset Time (VOT), and vowel fundamental frequency (F0) perturbation were quantified. Aspirated root onsets showed mean VOTs of 68-85 ms, while unaspirated root onsets averaged 12-22 ms. The prefix vowel duration showed compensatory shortening when preceding aspirated roots, indicating a prosodic foot-level timing constraint.

Tangkhul Linguistic Data & Analysis:

*** khe-sa [khasa] (to be hot) : VOT = 18 ms (root); Prefix aspiration = 45 ms**

Analysis: Prefix aspiration licensed before fricatives.

*** ke-kham [kakham] (to stop/hinder) : VOT = 78 ms (root); Prefix aspiration = 0 ms**

Analysis: Strict OCP dissimilation.

3. THEORETICAL PHONOLOGICAL IMPLICATIONS

These findings demonstrate that Tangkhul prefixal alternation is not mere morphological suppletion, but an active phonological rule governed by laryngeal licensing. It provides one of the few well-documented non-Indo-European cases of Grassmann's Law in the Sino-Tibetan language family.

Tangkhul Linguistic Data & Analysis:

*** ke-that [kathat] : to kill**

Analysis: PTB *g-sat; aspiration conditioned by ancient sibilant prefix.

REFERENCES & BIBLIOGRAPHY

* Gopal, Deepthi. 2016. Aspiration ’dissimilation’ in Tangkhul Naga. CMU Working Papers in Linguistics 8: 45-68.

* Ladefoged, Peter & Ian Maddieson. 1996. The Sounds of the World’s Languages. Blackwell.

* McCarthy, John. 1986. OCP Effects: Gemination and Antigemination. Linguistic Inquiry 17(2): 207-263.

Digital Archiving Notice:
Digitized and integrated into the Tangkhul Language Library Open Research Corpus.

TOWARDS MORPHOLOGICAL PROCESSING AND COMPUTATIONAL ANALYSI

Authors: Richard Luikham, Dr. Ninghorla Zimik, Dr. Kh. Raju Singh
Journal/Venue: Computational Linguistics & Natural Language Processing Journal | Year: 2020
Subject: Computational Linguistics & NLP | Language: English & Tangkhul

ABSTRACT:

Building language technology for indigenous, low-resource languages requires robust morphological analyzers and clean corpora. In this paper, we develop a computational pipeline for the Tangkhul language spoken in Manipur, India. Using a digitalized corpus of 150,000 words extracted from curriculum textbooks and folklore anthologies, we analyze token frequency distributions, morpheme boundaries, and syntactic function words. We implement a rule-enhanced finite-state segmenter that handles complex affixation, achieving an F1-score of 88.4% on out-of-vocabulary root identification and lemmatization.

Keywords: NLP, Morphological Segmentation, Tangkhul, Tokenization, Low-Resource NLP, Corpus

1. LINGUISTIC BACKGROUND & DIGITAL RESOURCE GAP

Despite possessing over 180,000 speakers and a century of published literature in Roman script, Tangkhul remains severely under-represented in modern computational language resources. Standard tools like spaCy and NLTK fail on Tangkhul text due to agglutinative suffix chains and unique function particles. We address this resource gap by compiling a standardized tokenized corpus and establishing baseline segmentation benchmarks.

Tangkhul Linguistic Data & Analysis:

* **eina : instrumental / ablative / causal postposition**
  Analysis: Most frequent particle (rel. freq: 4.8%).
* **kala : coordinating conjunction (and)**
  Analysis: Sentence and clause connector (rel. freq: 3.9%).
* **hi / chi : proximal ('this') / distal ('that') demonstrative**
  Analysis: Definiteness marker (rel. freq: 5.2%).

2. AFFIX SEGMENTATION ARCHITECTURE

Tangkhul verbs inflect through sequential agglutination: [Prefix] + Root + [Aspect/Valency] + [Tense/Mood] + [Clause Linker]. For example, the surface word 'i-thei-ser-ra-kaji' decomposes into root thei 'see', completive -ser, future -ra, and quotative -kaji. Our finite-state transducer segments these affix clusters systematically, resolving morphophonemic boundary adjustments.

Tangkhul Linguistic Data & Analysis:

* **sa-sak-kalu (sa-sak-ka-lu) : cause to make! (imperative causative)**
  Analysis: Root + Causative + Imperative.
* **ngashan-bing-wui (ngashan-bing-wui) : of the clans (genitive plural)**
  Analysis: Noun + Plural + Genitive.

3. EVALUATION & NLP APPLICATIONS

We evaluated our morphological analyzer on a gold-standard annotated test set of 2,500 sentences. The system achieved 91.2% precision and 85.8% recall on boundary detection. We demonstrate applications in bilingual dictionary search, computer-assisted language learning (CALL), and keyword extraction for digital heritage repositories.

Tangkhul Linguistic Data & Analysis:

*** kasakakhui (ka-sa-ka-khui) : compilation, gathering together**

Analysis: Double nominalized compound.

REFERENCES & BIBLIOGRAPHY

* Jurafsky, Daniel & James H. Martin. 2020. Speech and Language Processing (3rd ed. draft).

* Luikham, Richard, Ninghorla Zimik & Kh. Raju Singh. 2020. Towards the Study of Morphological Processing of the Tangkhul Language. arXiv:2009.07124.

* Tangkhul Literature Society. 2021. Tangkhul Grammar and Composition. Imphal: TLS Manipur.

Digital Archiving Notice:
Digitized and integrated into the Tangkhul Language Library Open Research Corpus.

ECHO WORD FORMATION AND REDUPLICATIVE MORPHOLOGY IN TANGKHU

Authors: Dr. S. Arokianathan

Journal/Venue: Indian Linguistics (Journal of the Linguistic Society of India) | Year: 1984
Subject: Descriptive Linguistics & Expressive Morphology | Language: English & Tangkhul

ABSTRACT:

In South Asian and Tibeto-Burman languages, reduplication and echo word formation constitute vital expressive devices imparting vivid sensory imagery and grammatical plurality. This paper investigates the phonological rules and semantic categories of reduplication in Tangkhul Naga. We distinguish between complete lexical reduplication (e.g., zingsho-zingsho 'daily'), partial reduplication with vowel alternation (e.g., chir-char 'bustling commotion'), and rhyming echo compounds that denote generalized collectives ('X and associated items').

Keywords: Reduplication, Echo Words, Expressives, Ideophones, Arokianathan, Indian Linguistics

1. MORPHOLOGICAL TYPES OF REDUPLICATION

Tangkhul reduplication operates at both inflectional and expressive levels. Type I consists of complete lexical repetition where a nominal or temporal base is duplicated to signify distribution or intensity. Type II consists of partial reduplication where the onset consonant is copied while the nucleus vowel or coda undergoes apophonic alternation (often high vowel to low vowel).

Tangkhul Linguistic Data & Analysis:

*** zingsho-zingsho : day by day, every day**

Analysis: Complete nominal reduplication of zingsho (day).

*** kachut-kachut : step-by-step, very slowly**

Analysis: Manner adverb reduplication.

*** ngam-ngam : sweetly, pleasantly fragrant**

Analysis: Sensory expressive reduplication.

2. ECHO WORDS AND GENERALIZED COLLECTIVES

Unlike Dravidian and Indo-Aryan languages which use a fixed consonant (such as gi- in Tamil or v- in Hindi) for echo formation, Tangkhul constructs echo compounds through semantic pairings of near-synonyms or coordinate clan items. The compound signifies not merely the literal items but the entire cultural domain or household equipment.

Tangkhul Linguistic Data & Analysis:

* **tui-tola (tui-tola) : words, rumours, talks, and gossip**
  Analysis: Echo pairing of tui (speech) with archaic tola.
* **shim-khor (shim-khor) : homestead and courtyard surroundings**
  Analysis: Spatial associative compound.
* **kachon-kachawui (kachon-kachawui) : clothing, garments and attire in general**
  Analysis: Textile collective noun.

3. IDEOPHONES AND ACOUSTIC ICONICITY

Tangkhul possesses an exceptionally rich inventory of ideophones depicting water movement, walking rhythms, and emotional states. These ideophones frequently feature reduplicated labials and liquid codas: hu-hu (rushing wind), chir-chir (running stream water), and phur-phur (rapid heart palpitation in fear).

Tangkhul Linguistic Data & Analysis:

* **hu-hu : sound of tempestuous wind blowing across peaks**
  Analysis: Onomatopoeic wind descriptor.
* **chir-chir : babbling sound of clear mountain spring water**
  Analysis: Aquatic sensory ideophone.
* **tek-tek : dripping of monsoon rain drops**
  Analysis: Precipitation acoustic iconicity.

REFERENCES & BIBLIOGRAPHY

* Abbi, Anvita. 1992. Reduplication in South Asian Languages: An Areal, Typological and Historical Study. Allied Publishers.

* Arokianathan, S. 1984. Echo Word Formation in Tangkhul Naga. Indian Linguistics 45: 33-46.

* Arokianathan, S. 1987. Tangkhul Naga Grammar. CIIL Grammar Series No. 16. Mysore: CIIL.

Digital Archiving Notice:
Digitized and integrated into the Tangkhul Language Library Open Research Corpus.

TONE SYSTEM AND PITCH ACCENT IN STANDARD UKHRUL TANGKHUL

Authors: Dr. Kapesa Khayi, Prof. K. V. Subbarao
Journal/Venue: Language in India & NEILS Linguistics Bulletin | Year: 2012
Subject: Phonetics & Tonology | Language: English & Tangkhul

ABSTRACT:

This paper investigates the phonetic and phonological properties of the tone system of Standard Ukhrul Tangkhul. While early missionary grammars treated Tangkhul as non-tonal or merely noted pitch variation, instrumental F0 analysis of monosyllabic minimal triplets reveals three distinct contrastive lexical tones: Tone 1 (High Level, 55), Tone 2 (Mid Level, 33), and Tone 3 (Low Falling, 21). We examine tone perturbation caused by syllable-final consonants and document tone sandhi processes occurring in nominal compounding and verbal suffixation.

Keywords: Tone, Pitch Accent, Tone Sandhi, Ukhrul, Fundamental Frequency, Khayi, NEILS

1. LEXICAL TONE CONTRASTS: MINIMAL TRIPLETS

The existence of tone in Tangkhul is conclusively demonstrated by numerous minimal triplets where segmental phonemes are identical but pitch contour differentiates semantic meaning. For instance, the monosyllabic frame /thang/ carries three distinct lexical meanings depending on whether it is uttered with High (55), Mid (33), or Low (21) pitch.

Tangkhul Linguistic Data & Analysis:

* **thang (Tone 1: High Level 55) : day, daylight period**
  Analysis: F0 fundamental frequency ~ 210 Hz.
* **thang (Tone 2: Mid Level 33) : to lift up, raise high**
  Analysis: F0 fundamental frequency ~ 165 Hz.
* **thang (Tone 3: Low Falling 21) : knife, machete, dao**
  Analysis: F0 fundamental frequency ~ 130 Hz.

2. TONAL ACOUSTIC MEASUREMENTS & F0 CONTOURS

Pitch contours were tracked across 10 native speakers (5 male, 5 female). Tone 1 exhibits a sustained flat high contour occupying the upper 20% of the speaker's vocal range. Tone 2 runs smoothly along the middle register. Tone 3 exhibits a slight initial rise followed by a steep descent, particularly before pause or sentence boundaries. Sonorant codas carry the pitch contour smoothly, whereas stop codas trigger abrupt glottalization that truncates the contour.

Tangkhul Linguistic Data & Analysis:

*** ka (High) vs ka (Mid) vs ka (Low) : to climb vs to bite vs to dry up**

Analysis: Monosyllabic vowel minimal set.

*** sØng (High) vs seng (Mid) vs sŁng (Low) : clean vs liver vs iron tool**

Analysis: Sonorant coda minimal set.

3. TONE SANDHI IN COMPOUNDING

When two monosyllabic words combine into a compound noun, the tone of the second element frequently undergoes sandhi. A Low tone following a High tone is raised to Mid to prevent extreme pitch leaps within a single phonological word. These tone sandhi rules operate systematically across both spoken discourse and traditional musical poetry.

Tangkhul Linguistic Data & Analysis:

* **shim (Mid) + khor (Low) -> shimkhor [33-33] : courtyard**
  Analysis: Low tone raising following Mid stem.
* **lui (High) + ram (Low) -> luiram [55-33] : cultivated field territory**
  Analysis: Low tone raising following High stem.

REFERENCES & BIBLIOGRAPHY

* Khayi, Kapesa & K. V. Subbarao. 2012. Tone System and Pitch Accent in Standard Ukhrul Tangkhul. Language in India 12(8): 211-235.

* Maddieson, Ian. 1978. Universals of Tone. In Greenberg (ed.), Universals of Human Language, Vol. 2, 335-365.

* Yip, Moira. 2002. Tone. Cambridge Language Surveys. Cambridge University Press.

Digital Archiving Notice:
Digitized and integrated into the Tangkhul Language Library Open Research Corpus.

TANGKHUL LUIRA PHANIT (SEED SOWING FESTIVAL) AND ANCIENT AGRICU

Authors: Prof. Khashim Ruivah

Journal/Venue: Journal of North East India Council for Social Science Research & TRI Manipur | Year: 1987
Subject: Ethnolinguistics, Cultural Anthropology & Ritual Chants | Language: English & Tangkhul

ABSTRACT:

Luira Phanit (the Seed Sowing Festival) is the supreme cultural and agricultural festival of the Tangkhul Naga people, celebrated annually in February across all traditional villages. This paper provides an ethnolinguistic documentation of the ceremonial liturgy performed during Luira. We record and translate the sacred Chithar (first seed-sowing invocation) chanted exclusively by the village chief (Awungshi), the liturgical formulas for domestic hearth renewal, and the poetic lyrics of the Hao Pheichak (traditional circle dance). The texts preserve archaic lexical roots unrecorded in modern dictionaries.

Keywords: Luira Phanit, Chithar, Awungshi, Hao Pheichak, Agrarian Liturgy, Ruivah, Manipur

1. CULTURAL SIGNIFICANCE & CHRONOLOGY OF LUIRA

Luira Phanit marks the dawn of the agricultural calendar. Before any villager may sow seed in their terraced fields, the village priest and chief (Awungshi) must perform the sacred ritual of Chithar (first seed offering) to conciliate Ameo (the benevolent earth guardian deity). The festival extends across seven ceremonial days: Luishat (preparation), Phanit-khara (arrival), Pheichak (communal dance), and Ngashan (clan feasting).

Tangkhul Linguistic Data & Analysis:

* **Luira Phanit : Seed sowing festival**
  Analysis: lui (field) + ra (seed/root) + phanit (sacred feast).
* **Chithar : first sacred seed sowing ritual**
  Analysis: Chi (paddy seed) + thar (consecration/first blessing).
* **Awungshi : hereditary village chief and high priest**
  Analysis: Custodian of village agrarian rituals.

2. SACRED INVOCATIONAL LITURGY: THE CHITHAR PRAYER

At dawn on the third day, the Awungshi enters the ancestral sacred plot (Awung-luiram) accompanied by clan elders. Scattering seven handfuls of consecrated paddy seeds onto the newly turned earth, he chants the ancient invocation pleading for abundant rain, pest protection, and communal bumper harvests.

Tangkhul Linguistic Data & Analysis:

* **’Kazingkhamang eina rarlu, Kazingramwui tui-ngapom hung-kachamlaga ngaleili hung-sungsaklu!’ : O celestial rains come forth,**
  Analysis: Archaic meteorological invocation.
* **’Hao naorapli kasham makasalu, theiwon kapha khangaraorap eina kashungsaklu!’ : Spare the children of Hao from famine, fill th**
  Analysis: Community protection formula.

3. THE HAO PHEICHAK DANCE & POETIC ANTIPHONY

The festival culminates in the Hao Pheichak dance performed in the village square (Mungchan). Dressed in ceremonial woven shawls (Haora and Luirim) and feather headdresses, young men and women sing antiphonal call-and-response verses honoring ancestral lineages, romantic longing, and agricultural fertility.

Tangkhul Linguistic Data & Analysis:

* **Haora kala Luirim : Sacred ceremonial male and female woven shawls**
  Analysis: Traditional tribal textile regalia.
* **Mungchan : sacred central village festival plaza**
  Analysis: Civic ritual arena.

REFERENCES & BIBLIOGRAPHY

* Hodson, T. C. 1911. The Naga Tribes of Manipur. London: Macmillan and Co.

* Luikham, T. 1961. Wung Tangkhul Naga Okthot Mayonza. Imphal.

* Ruivah, Khashim. 1987. Tangkhul Luira Phanit and Ancient Agricultural Invocations. JNEICSSR 11(2): 15-38.

Digital Archiving Notice:
Digitized and integrated into the Tangkhul Language Library Open Research Corpus.

TANGKHUL HAO NGASHAN: CUSTOMARY LAW, CLAN EXOGAMY, AND SOCIA

Authors: T. Luikham, A. S. W. Shimray, Wung Cultural Research Forum
Journal/Venue: Tribal Jurisprudence & Cultural Bulletin of Manipur | Year: 2004
Subject: Customary Law, Tribal Jurisprudence & Clan Governance | Language: English & Tangkhul

ABSTRACT:

Hao Ngashan embodies the unwritten constitutional and customary legal code governing Tangkhul society since time immemorial. This paper analyzes the structural pillars of Tangkhul customary jurisprudence: the patrilineal clan system (Shang), strict clan exogamy (Kashan), land tenure mechanisms, the judicial authority of the Village Council (Hangva), and traditional dispute settlement procedures. We examine the sacred oath-taking formula (Tuishot) and demonstrate how customary law operates harmoniously alongside modern constitutional protections in Manipur.

Keywords: Hao Ngashan, Customary Law, Clan Exogamy, Hangva, Shimxang, Tuishot, Jurisprudence

1. THE CLAN (SHANG) AND EXOGAMOUS MARRIAGE (KASHAN)

Tangkhul social structure is organized into patrilineal clans (Shang) that trace their descent from common ancestors. The bedrock principle of social ethics is Kashan--strict clan exogamy. Intermarriage within the same Shang is considered an intolerable moral taboo (Kashansha) that historically incurred banishment from the village territory. All marriages must link distinct Shangs, creating enduring reciprocal obligations and alliances across the community.

Tangkhul Linguistic Data & Analysis:

* **Shang : patrilineal clan, lineage house**
* **Kashan : customary law of clan exogamy**

  Analysis: Fundamental marital taboo and governance rule.
* **Kashansha : incestuous breach of clan exogamy**

  Analysis: Social transgression incurring ostracism.

2. THE VILLAGE COURT (HANGVA) & JUDICIAL ADMINISTRATION

Civic, property, and criminal disputes are adjudicated by the Hangva (Village Council and Court of Elders), presided over by the hereditary Awungshi (chief). Each major clan is represented by a senior elder (Hangva-khana). The court prioritizes reconciliation (Tui-ngalung) and restitution rather than retributive punishment. Fines historically took the form of cattle, swine, woven blankets (Haora), or rice beer (Khor).

Tangkhul Linguistic Data & Analysis:

* **Hangva : village council of clan elders / traditional court**
  Analysis: Judicial bench of the village.
* **Tui-ngalung : peaceful reconciliation, amicable arbitration**
  Analysis: Consensus-based restorative justice.
* **Khor : traditional rice brew used in customary treaty sealing**
  Analysis: Sacred drink sealing reconciled pacts.

3. THE SACRED OATH FORMULA: TUISHOT

When evidence in boundary or theft disputes is ambiguous and witnesses are divided, the dispute is resolved by Tuishot (the solemn oath before earth and heaven). The disputing parties bite a tiger tooth (Haram-ha) or plunge stones into sacred water, invoking divine retribution upon whoever utters falsehood. The fear of ancestral and cosmic retribution ensured near-absolute truthfulness in customary legal proceedings.

Tangkhul Linguistic Data & Analysis:

*** Tuishot : solemn traditional oath-taking ordeal**

Analysis: Supreme evidentiary oath before spirits and earth.

*** Haram-ha : tiger tooth used in sacred oath trials**

Analysis: Symbol of wild retribution against perjury.

REFERENCES & BIBLIOGRAPHY

* Luikham, T. 1961. Wung Tangkhul Naga Okthot Mayonza. Imphal: Tangkhul Literature Society.

* Shimray, A. S. W. 2001. History of the Tangkhul Nagas. New Delhi: Akansha Publishing House.

* Wung Cultural Research Forum. 2004. Tangkhul Hao Ngashan: Customary Law and Jurisprudence. Imphal.

Digital Archiving Notice:
Digitized and integrated into the Tangkhul Language Library Open Research Corpus.

TANGKHUL KINSHIP TERMINOLOGY, CONSANGUINITY, AND LINEAGE STRUC

Authors: Prof. M. S. Thirumalai, Dr. Kapesa Khayi
Journal/Venue: Language in India (ISSN 1930-2940) | Year: 2014
Subject: Ethnolinguistics & Kinship Anthropology | Language: English & Tangkhul

ABSTRACT:

Kinship terminologies embody the social structural logic and reciprocal obligations of human societies. This paper presents the first exhaustive ethnolinguistic classification of Tangkhul kinship terminology, encompassing 68 consanguineal and affinal terms. We demonstrate that Tangkhul exhibits a bifurcate merging system of the Omaha subtype, reflecting strict patrilineal clan organization (Shang) and asymmetrical cross-cousin marriage rules.

Keywords: Kinship, Omaha Type, Shang, Consanguinity, Ethnolinguistics, Thirumalai

1. BIFURCATE MERGING & PATRILINEAL ALIGNMENT

In Tangkhul, the father’s brother is addressed by the same root as father (Ava-kakhano for father’s younger brother, Ava-kakhara for father’s elder brother), whereas the mother’s brother is distinguished as Mawa/Kuku. Similarly, parallel cousins are equated with biological siblings (Uningnao), whereas cross-cousins fall into distinct terminological classes linked to marriageability.

Tangkhul Linguistic Data & Analysis:

*** Ava : father**

Analysis: Primary consanguineal male ancestor.

*** Ava-kakhano : father’s younger brother**

Analysis: Bifurcate merging parallel uncle.

*** Mawa / Kuku : mother’s brother (maternal uncle)**

Analysis: Affinal alliance partner.

2. SIBLING TERMINOLOGY & SENIORITY DISTINCTIONS

Seniority relative to the speaker is strictly encoded in sibling terms. Elder brother is achon/ayei, younger brother is atat/akhanao, elder sister is va-kakhara, and younger sister is va-kakhano. The collective compound uningnao encompasses all sibling kin within the Shang lineage.

Tangkhul Linguistic Data & Analysis:

*** ayei : elder brother**

Analysis: Senior male sibling.

*** akhanao : younger sibling (male or female)**

Analysis: Junior sibling term.

*** uningnao : lineage siblings / clan brethren**

Analysis: Collective kinship noun.

3. AFFINAL TERMINOLOGIES & ASYMMETRICAL ALLIANCES

Affinal terms reflect mother’s brother’s daughter (MBD) preferential alliance. The wife’s clan is honored as Mayang-shang (life-giving in-laws), commanding ceremonial gifts of meat and woven shawls at all lifecycle rituals.

Tangkhul Linguistic Data & Analysis:

*** Mayang-shang : wife-giving affinal clan lineage**

Analysis: Ceremonial in-law alliance.

REFERENCES & BIBLIOGRAPHY

* Fox, Robin. 1967. Kinship and Marriage: An Anthropological Perspective. Penguin.

* Murdock, George Peter. 1949. Social Structure. Macmillan.

* Thirumalai, M. S. & Kapesa Khayi. 2014. Tangkhul Kinship Terminology. Language in India 14(5): 112-138.

Digital Archiving Notice:

Digitized and integrated into the Tangkhul Language Library Open Research Corpus.

GRAMMATICALIZATION OF MOTION VERBS INTO DIRECTIONAL POSTPOSITI

Authors: Dr. Rekha Sharma, Dr. S. Arokianathan
Journal/Venue: Linguistics of the Tibeto-Burman Area | Year: 2008
Subject: Syntax, Semantics & Grammaticalization | Language: English & Tangkhul

ABSTRACT:

This paper investigates the grammaticalization pathways through which full lexical motion verbs in Tangkhul have developed into directional markers, telic aspectual affixes, and spatial postpositions. Drawing on connected texts spanning a century of documentation, we demonstrate that motion verbs specifying vertical axis (ascent/descent) and deictic orientation (centrifugal vs centripetal) systematically fuse with verbal stems to form complex path predicates.

Keywords: Grammaticalization, Motion Verbs, Directionals, Deixis, Spatial Grammar, LTBA

1. LEXICAL SOURCE VERBS & DEICTIC TRAJECTORIES

Tangkhul distinguishes three basic deictic path vectors: (i) Centripetal movement toward speaker (-ung eina ungkaphung), (ii) Centrifugal movement away from speaker (-va eina vazatkasa), and (iii) Altitude-conditioned vectors: ascent uphill (-ka) and descent downhill (-tuk).

Tangkhul Linguistic Data & Analysis:

* **ung-kaphung : to arrive coming home toward speaker**
  Analysis: Centripetal motion.
* **va-zatkasa : to go away departing from speaker**
  Analysis: Centrifugal motion.
* **thuk-kashok : to climb up out of a gorge onto peak**
  Analysis: Ascending vertical vector.

2. VERBAL SERIALISATION & AFFIXATION

In serial verb constructions, the directional verb loses its independent nominal arguments and undergoes phonological reduction into an aspectual clitic. For instance, in ’a ot sa-va-ye’ (He went off to do work), the auxiliary -va marks departure aspect rather than physical walking.

Tangkhul Linguistic Data & Analysis:

*** sa-va-ye : did and departed / went off doing**

Analysis: Grammaticalized departure clitic.

*** thei-shok-e : perceived clearly outward / manifested**

Analysis: Directional verb shok (exit) grammaticalized as telic resultative.

3. TYPOLOGICAL PARALLELS IN TIBETO-BURMAN

The Tangkhul grammaticalization cline [Motion Verb -> Serial Verb -> Directional Affix -> Aspect Marker] directly mirrors paradigms observed in Qiang, Jingpho, and Lahu, providing evidence for areal Tibeto-Burman syntactic evolution.

Tangkhul Linguistic Data & Analysis:

*** kashok (exit) -> -shok (telic completive) : Grammaticalization pathway**

Analysis: Lexical verb to grammatical morpheme.

REFERENCES & BIBLIOGRAPHY

* Bybee, Joan, Revere Perkins & William Pagliuca. 1994. The Evolution of Grammar. University of Chicago Press.

* Heine, Bernd & Tania Kuteva. 2002. World Lexicon of Grammaticalization. Cambridge University Press.

* Sharma, Rekha & S. Arokianathan. 2008. Grammaticalization of Motion Verbs in Tangkhul. LTBA 31(2): 67-89.

Digital Archiving Notice:
Digitized and integrated into the Tangkhul Language Library Open Research Corpus.

OPTIMALITY THEORETIC ACCOUNT OF TONE SANDHI IN TANGKHUL NOMINA

Authors: Dr. Kapesa Khayi

Journal/Venue: Phonological Studies of the North East (NEILS Occasional Papers) | Year: 2015
Subject: Theoretical Phonology & Optimality Theory | Language: English & Tangkhul

ABSTRACT:

When monosyllabic morphemes concatenate into compound nouns in Tangkhul Naga, systematic tonal alternations occur to avoid adjacent sharp pitch falls within the prosodic word. This paper formulates an Optimality Theoretic (OT) analysis of Tangkhul tone sandhi. We show that the interaction of markedness constraints (*NoExtremeDrop, OCP-Tone) ranked above faithfulness constraints (Ident-IO(Tone)) successfully accounts for right-edge register raising.

Keywords: Optimality Theory, Tone Sandhi, Phonology, Constraints, Ukhrul, F0

1. THE EMPIRICAL PROBLEM OF TANGKHUL SANDHI

Standard Ukhrul Tangkhul possesses three tones: High (H, 55), Mid (M, 33), and Low (L, 21). When a High-toned base compounds with a Low-toned nominal root, the surface output is consistently High-Mid (H-M) rather than the expected H-L. For example, lui (H, field) + ram (L, land) surfaces as [lui-ram] with Mid pitch on the second syllable.

Tangkhul Linguistic Data & Analysis:

*** lui [55] + ram [21] -> lui-ram [55-33] : cultivated field territory**

Analysis: Right-edge Low raising to Mid.

*** thang [55] + zingsho [33] -> thang-zingsho [55-33] : broad daylight**

Analysis: Faithful Mid retention.

2. OT CONSTRAINT HIERARCHY

We propose the constraint ranking: *Drop-HL >> OCP(Tone) >> Ident-IO(Tone). The high-ranked markedness constraint *Drop-HL prohibits a two-register pitch plunge (55 to 21) within a single prosodic word. Candidate [55-33] incurs a single violation of low-ranked faithfulness Ident-IO(Tone), but decisively eliminates the faithful candidate [55-21].

Tangkhul Linguistic Data & Analysis:

* **Candidate A: [55-21] : Violates *Drop-HL**
  Analysis: Fatal violation on markedness.
* **Candidate B: [55-33] : Optimal winner**
  Analysis: Preserves optimal prosodic contour.

3. COMPOUND TRUNCATION AND CODA EFFECTS

Syllables with glottal stop codas inhibit sandhi raising, behaving as opaque phonological boundaries. This demonstrates that laryngeal features interact directly with tonal autosegmental licensing in Tangkhul phonology.

Tangkhul Linguistic Data & Analysis:

*** Glottal opacity : Coda [ʔ] blocks tone assimilation**

Analysis: Laryngeal feature licensing.

REFERENCES & BIBLIOGRAPHY

* Khayi, Kapesa. 2015. Tone Sandhi in Tangkhul Compounding. NEILS Occasional Papers 4: 145-168.

* Prince, Alan & Paul Smolensky. 2004. Optimality Theory: Constraint Interaction in Generative Grammar. Blackwell.

* Yip, Moira. 2002. Tone. Cambridge University Press.

Digital Archiving Notice:
Digitized and integrated into the Tangkhul Language Library Open Research Corpus.

SYLLABLE WEIGHT AND METRICAL STRUCTURE IN UKHRUL TANGKHUL

Authors: Prof. P. Madhubala, Dr. Ninghorla Zimik

Journal/Venue: Journal of the Linguistic Society of North East India | Year: 2017
Subject: Metrical Phonology & Prosody | Language: English & Tangkhul

ABSTRACT:

This paper examines the relationship between syllable weight, moraic structure, and stress assignment in Standard Ukhrul Tangkhul. While Tangkhul has traditionally been described as a tone language without prominent stress, acoustic analysis reveals that primary metrical stress consistently falls upon the final heavy syllable (bimoraic ??) of the phonological stem, relegating light prefixal syllables (monomoraic ?) to metrically weak prosodic positions.

Keywords: Syllable Weight, Moraic Theory, Metrical Feet, Stress, Ukhrul, Tangkhul

1. SYLLABLE TYPOLOGY AND MORA COUNT

Tangkhul syllables conform to the maximal template (C)(C)V(C). Syllables ending in vowels (CV) are monomoraic (light, ?), while syllables closed by sonorant or stop codas (CVC) and syllables with diphthongs (CVV) are bimoraic (heavy, ??). Only bimoraic syllables can carry the full three-way tonal contrast in citation form.

Tangkhul Linguistic Data & Analysis:

*** ke- (prefix) : Light syllable (?)**

Analysis: Unstressed, reduced vowel.

*** -pha:ng (stem) : Heavy syllable (??)**

Analysis: Stressed stem carrying lexical tone.

2. METRICAL FOOT BINARITY & IAMBIC GROUPING

Disyllabic prefix-stem words form canonical iambic feet (L H), where the light prefix serves as the weak foot member and the heavy root serves as the rhythmic head. This explains why prefix vowels are phonetically reduced to schwa [e].

Tangkhul Linguistic Data & Analysis:

*** [ke.that] : to kill (iambic foot: L H)**

Analysis: Primary stress on heavy root.

*** [me.thØj] : does not see (iambic foot: L H)**

Analysis: Negative prefix schwa reduction.

3. RHYTHMIC IMPLICATIONS FOR FOLK VERSE

Traditional Tangkhul oral poetry (Hao-laa) exploits this bimoraic foot structure. Chants are organized into four-mora hemistichs, establishing a steady musical pulse that coordinates choral dancing during festivals.

Tangkhul Linguistic Data & Analysis:

* **Hao-laa metrical meter : Four-mora rhythmic cadence**

  Analysis: Prosodic coordination in dance.

REFERENCES & BIBLIOGRAPHY

* Hayes, Bruce. 1995. Metrical Stress Theory: Principles and Case Studies. University of Chicago Press.

* Madhubala, P. & Ninghorla Zimik. 2017. Syllable Weight in Tangkhul. JLSNEI 6(1): 55-76.

Digital Archiving Notice:
Digitized and integrated into the Tangkhul Language Library Open Research Corpus.

ETHNOBOTANICAL WISDOM AND TRADITIONAL HERBAL MEDICINE OF THE T

Authors: Dr. Ringkahao Horam, Dr. V. S. Zimik

Journal/Venue: Indian Journal of Traditional Knowledge (IJTK / CSIR-NIScPR) | Year: 2019
Subject: Ethnobotany, Traditional Medicine & Ecology | Language: English & Tangkhul

ABSTRACT:

The biodiverse montane forests of Ukhrul and Kamjong districts harbor an extraordinary repository of botanical species. This paper documents the traditional ethnobotanical knowledge of the Tangkhul Nagas based on ethnopharmacological fieldwork with 14 practicing village herbal healers (Khangashun-Kaphanao). We catalogue 85 medicinal plant species across 42 families, detailing native Tangkhul nomenclature, plant parts used, preparation modes, and therapeutic applications for fevers, bone fractures, digestive ailments, and poison antidotes.

Keywords: Ethnobotany, Herbal Medicine, Phytotherapy, Ukhrul, Shirui, Zimik, IJTK

1. NATIVE BOTANICAL TAXONOMY (TANGKHUL PLANT CLASSIFICATION)

Tangkhul folk taxonomy organizes the plant kingdom into major morphological life-forms: Thingphun (large standing timber trees), Wonra/Tingra (climbing lianas and woody vines), Shinao/Khangaphei (herbaceous forest floor plants), and Thingthei (fruit-bearing flora). Medicinal efficacy is categorized according to taste qualities: kakhui (bitter/antipyretic), kasha (pungent/anti-inflammatory), and ngam-kasa (sweet/tonic).

Tangkhul Linguistic Data & Analysis:

* **Thingphun : large standing tree class**
* **Tingra : climbing medicinal lianas / creeping vines**
* **kakhui : bitter tasting therapeutic property**

2. KEY MEDICINAL SPECIES & INDIGENOUS PREPARATIONS

Foremost among Tangkhul therapeutic herbs is Zingra (Houttuynia cordata), used raw for digestive and antimicrobial cures. The bark of Thing-kakhui (Mahonia manipurensis) is boiled into an intense yellow tonic for jaundice and malaria. Fresh poultices of Won-khamatha (Clerodendrum colebrookianum) are applied to regulate hypertension and cardiac distress.

Tangkhul Linguistic Data & Analysis:

* **Zingra : Houttuynia cordata (medicinal fish mint)**
  Analysis: Digestive leaf antibiotic.
* **Thing-kakhui : Mahonia manipurensis (golden bark barberry)**
  Analysis: Bark decoction for fevers and jaundice.
* **Meina-kasha : Zanthoxylum armatum (winged prickly ash / wild Sichuan pepper)**
  Analysis: Dental pain and stomachic tonic.

3. CONSERVATION ETHICS & SACRED GROVES (KHAMASUNG)

Harvesting medicinal plants is governed by customary taboos (Khamasung). Healers never uproot entire plant colonies, harvesting only mature leaves or lateral roots while chanting prayers of apology to the forest spirit (Luiram Ameo). This ecological ethic preserved rare montane herbs for centuries.

Tangkhul Linguistic Data & Analysis:

*** Khamasung : sacred conservation taboo and harvesting ethics**

Analysis: Traditional ecological sanctuary rule.

REFERENCES & BIBLIOGRAPHY

* Horam, Ringkahao & V. S. Zimik. 2019. Ethnobotanical Wisdom of Tangkhul Nagas. IJTK 18(3): 412-428.

* Jain, S. K. 1991. Dictionary of Indian Folk Medicine and Ethnobotany. Deep Publications.

Digital Archiving Notice:
Digitized and integrated into the Tangkhul Language Library Open Research Corpus.

TANGKHUL FOLK RIDDLES (TUNGTUKTUI): METAPHORICAL COGNITION AND

Authors: Prof. Khashim Ruivah, Dr. Kapesa Khayi
Journal/Venue: Folklore & Oral Traditions of Northeast India | Year: 2011
Subject: Folklore, Oral Riddles & Cognitive Linguistics | Language: English & Tangkhul

ABSTRACT:

Tungtuktui (folk riddles) constitute one of the most vibrant intellectual pastimes of traditional Tangkhul evening gatherings in village dormitories and family hearths. This paper investigates the cognitive structures, metaphor mappings, and pedagogical functions of Tangkhul riddling. We analyze 400 collected riddles across four semantic domains: human anatomy, household implements, flora/fauna, and celestial phenomena, demonstrating how riddles cultivate rapid analogical reasoning in youth.

Keywords: Tungtuktui, Riddles, Folklore, Metaphor, Oral Literature, Cognition

1. THE RIDDLING FRAME AND DISCOURSE CONVENTIONS

A riddling session opens with the ritual formula: 'Tungtuk-tui akha hang-sa!' (Let me cast a riddle!). The audience responds in unison: 'Rarlu, ithumna pamsaye!' (Bring it forth, we are ready!). The challenger poses the metaphoric enigma, and the participants offer competing guesses until the correct solution is hailed.

Tangkhul Linguistic Data & Analysis:

* **Tungtuk-tui akha hang-sa! : Formulaic opening: Let me cast a riddle!**
  Analysis: Discourse initiation marker.
* **Tungtuktui : folk riddle, metaphorical enigma**
  Analysis: Tungtuk (strike/probe) + tui (word).

2. STRUCTURAL TYPOLOGY: ANIMAL AND ANATOMY METAPHORS

Tangkhul riddles rely on vivid domestic comparisons. For instance, the traditional carrying basket (Phurwon) is personified as a woman who eats firewood and carries stones; teeth are thirty-two white warriors standing guard in a crimson cave; and rain is a thousand silver needles sewing sky to mountain.

Tangkhul Linguistic Data & Analysis:

* **'Mayarnaobing thakhara chon kachak sanglaga pamye: Khikha?' -> Kha (Teeth) : Thirty white-robed youths sitting in rows: What**
  Analysis: Anatomical personification.
* **'Zing eina tuitsang ngaleili vashukla ngashiman-serie: Khikha?' -> Kazingkhamik (Rain) : Falls from heaven, touches earth and va**
  Analysis: Meteorological riddle.

3. PEDAGOGICAL AND SOCIAL COGNITIVE VALUE

Beyond recreation, Tungtuktui served as oral encyclopedias. Children learned animal tracking habits, forest plant traits, and linguistic vocabulary by deciphering these poetic enigmas during winter evenings beside the glowing hearth.

Tangkhul Linguistic Data & Analysis:

*** Meikhangahui wari : fireside hearth education**

Analysis: Informal traditional pedagogy.

REFERENCES & BIBLIOGRAPHY

* Georges, Robert A. & Alan Dundes. 1963. Toward a Structural Definition of the Riddle. Journal of American Folklore.

* Ruivah, Khashim & Kapesa Khayi. 2011. Tangkhul Folk Riddles. Folklore & Oral Traditions 9: 88-115.

Digital Archiving Notice:
Digitized and integrated into the Tangkhul Language Library Open Research Corpus.

ORAL HISTORIOGRAPHY AND MIGRATION ROUTES OF THE TANGKHUL NAG

Authors: A. S. W. Shimray, Prof. Gangmumei Kamei
Journal/Venue: North East Indian Historical Association (NEIHA Proceedings) | Year: 2012
Subject: Oral History, Migration & Ethnogeography | Language: English & Tangkhul

ABSTRACT:

The prehistoric migrations and settlement trajectories of the Tangkhul Nagas are preserved through meticulously transmitted oral chants (Khararchan) and village founding genealogies. This paper correlates oral historiographical evidence with toponyms, megalithic trail markers, and ceramic archaeological remains across the mountain ridges of Manipur and Myanmar. We trace the dispersal from the sacred dispersal center of Makhel, the founding of Hundung, and the subsequent expansion into the central, northern, and eastern valleys of Ukhrul and Kamjong.

Keywords: Migration, Makhel, Hundung, Ukhrul, Oral History, NEIHA, Shimray

1. THE DISPERSAL FROM MAKHEL (THE SACRED PEAR TREE)

Universal Naga oral tradition identifies Makhel in northern Manipur as the ancestral dispersal point. According to the Tangkhul epic chant, three brothers (representing the Meithei of the valley, the Tangkhul of the east, and the western Nagas) parted company beside the sacred pear tree (Chiteikayung). Before parting, they swore an eternal fraternal pact, promising to reunite in peace and harmony.

Tangkhul Linguistic Data & Analysis:

* **Makhel : sacred ancestral dispersal cradle of the Nagas**
  Analysis: Cradle of highland migration.
* **Chiteikayung : sacred flowering pear tree of parting**
  Analysis: Megalithic covenant site.
* **uningnao mathat : fraternal kinship pact between brothers**
  Analysis: Foundational inter-tribal peace oath.

2. SETTLEMENT OF HUNDUNG & DIVISION OF TERRITORIES

Migrating eastward along mountain crests, the Tangkhuls established their primary settlement at Hundung. The elder brother remained at Hundung as chief, while younger lineages pushed northward to establish Ukhrul, Shirui, Phadang, and east toward the Somra tract across the Chindwin watershed.

Tangkhul Linguistic Data & Analysis:

*** Hundung : historic mother village of central Tangkhul territory**

Analysis: Primary dispersal seat in Ukhrul.

*** Awungshi Hundung : chief of Hundung, elder line of the migration**

Analysis: Hereditary custodian of the dispersal.

3. LINGUISTIC DIALECT BOUNDARIES AND TOPONYMIC MEMORY

The migration pathways correlate precisely with the five major dialect clusters of the Tangkhul language today: North (Tusom, Raphei), Central (Ukhrul), West (Phadang, Kachai), South (Bungpa, Kamjong), and East (Somra). Ancient village boundary names retain archaic vocabulary that confirms centuries of continuous highland habitation.

Tangkhul Linguistic Data & Analysis:

* **Raphei (Northern hills) / Somra (Eastern valley) : Geographical dialect sectors**

  Analysis: Regional toponymy.

REFERENCES & BIBLIOGRAPHY

* Kamei, Gangmumei. 2004. History of the Zeliangrong Nagas: From Makhel to Rani Gaidinliu. Spectrum.

* Shimray, A. S. W. 2001. History of the Tangkhul Nagas. New Delhi: Akansha.

* Shimray, A. S. W. & Gangmumei Kamei. 2012. Migration Routes of the Tangkhul Nagas. NEIHA Proceedings 32: 204-226.

Digital Archiving Notice:
Digitized and integrated into the Tangkhul Language Library Open Research Corpus.

TANGKHUL WEAVING AND TEXTILE SEMIOTICS: CULTURAL CODES OF HAO

Authors: Dr. Ninghorla Zimik, Prof. M. S. Thirumalai
Journal/Venue: Textile Anthropology & Cultural Studies Review | Year: 2016
Subject: Textile Anthropology, Weaving Semiotics & Material Culture | Language: English & Tangkhul

ABSTRACT:

Traditional handloom textiles constitute one of the most sophisticated visual languages and identity markers in Naga civilization. This paper analyzes the technology, color symbolism, and semiotic coding of traditional Tangkhul loin-loom (Porkapong) weaving. We examine the structural design, stripe frequencies, and ritual privileges associated with the male warrior shawl (Haora), the elite ceremonial wrap (Luirim), and the female wrap-around skirt (Kashan / Phingkham), demonstrating how textile motifs encode social status, clan affiliation, and cosmological harmony.

Keywords: Weaving, Haora, Luirim, Kashan, Loin Loom, Textile Semiotics, Zimik

1. LOOM TECHNOLOGY: THE TRADITIONAL LOIN LOOM (PORKAPONG)

Tangkhul weaving is executed exclusively by women on the indigenous backstrap or loin loom (Porkapong / Khashan-thing). The loom consists of lightweight bamboo and polished timber rods (Tingkhor, Saphur, and Machor) tensioned by the weaver's own body weight. This portable loom permits exceptionally dense warp-faced weaves capable of withstanding decades of highland wear.

Tangkhul Linguistic Data & Analysis:

* **Porkapong : traditional indigenous backstrap loin loom**
* **Tingkhor : bamboo warp-beam rod**
* **Machor : polished timber batten / beater sword**

2. COLOR SEMIOTICS AND STRUCTURAL MOTIFS: HAORA & LUIRIM

Color palettes are drawn from native natural dyes: Meithai-ra (wild madder) for deep crimson red, Kharong-thing (indigo leaves) for deep raven black, and unbleached cotton for virgin white. The Haora shawl features broad black and white longitudinal bands framed by red selvedges, symbolizing bravery, purity, and warrior defense. The Luirim shawl, featuring intricate geometric diamond motifs (Kongra won), was historically reserved for village royalty and chiefs.

Tangkhul Linguistic Data & Analysis:

* **Haora : traditional black and red striped male shawl**
  Analysis: Symbol of masculine dignity and defense.
* **Luirim : aristocratic ceremonial wrap with diamond motifs**
  Analysis: Reserved for chiefs and feast-givers.
* **Kongra won : diamond river-ripple geometric textile motif**
  Analysis: Aquatic cosmological emblem.

3. FEMALE ATTIRE: KASHAN AND PHINGKHAM

The Tangkhul woman’s wrap-around skirt (Kashan) serves as an overt visual manifesto of lineage identity. The Phingkham kashan, adorned with bold scarlet and black stripes, is worn during sacred festivals, whereas the Seichang kashan features delicate parallel lines worn during daily farm labor.

Tangkhul Linguistic Data & Analysis:

* **Phingkham kashan : sacred festive female wrap-around skirt**
  Analysis: Formal ceremonial attire.
* **Seichang kashan : everyday agrarian working wrap skirt**
  Analysis: Practical agricultural textile.

REFERENCES & BIBLIOGRAPHY

* Alemchiba, M. 1968. The Arts and Crafts of Nagaland. Naga Institute of Culture, Kohima.

* Zimik, Ninghorla & M. S. Thirumalai. 2016. Tangkhul Weaving and Textile Semiotics. Textile Anthropology Review 15(2): 142-168.

Digital Archiving Notice:
Digitized and integrated into the Tangkhul Language Library Open Research Corpus.